FinGuard AI — Problem Statement

Overview

Banks and NBFCs make three high-stakes, high-volume decisions every day — who gets a loan, which transactions are fraudulent, and whether a submitted identity document is genuine — while also fielding constant policy and compliance questions from their own staff. Today these are handled by separate, disconnected tools (rule-based credit checks, manual document review, static FAQ documents), which is slow, inconsistent, and hard to explain when regulators or customers ask "why was I rejected?"

FinGuard AI is a single platform that unifies all four of these into one system: a credit risk engine, a fraud detection engine, a document verification engine, and a conversational compliance assistant that can query the other three — all served through one deployed, monitored production pipeline.

Problem

1. **Credit decisions** are often opaque — applicants and regulators can't get a clear reason for rejection, and manual review doesn't scale.
2. **Transaction fraud** patterns shift constantly and are too subtle and fast-moving for static rule-based systems to catch reliably.
3. **KYC document verification** is manual, slow, and inconsistent across reviewers.
4. **Policy and compliance knowledge** is scattered across PDFs and internal wikis, so staff waste time searching instead of getting a direct, grounded answer — and no existing tool connects that knowledge to a specific applicant's actual risk data.

Objectives

- Predict credit risk for loan applicants with a model whose decisions can be explained in plain terms (per-applicant, not just in aggregate).
- Detect fraudulent transactions in near real time, tuned for a highly imbalanced dataset where fraud is a small fraction of all activity.
- Automatically verify and extract data from identity documents submitted during onboarding.
- Give compliance/ops staff a conversational assistant that answers policy questions using real internal documents, and can pull live risk/fraud/verification results into its answers on request.
- Package all of this as containerized services behind one gateway, with model versioning and drift monitoring — not just notebooks that run once.

Target users

- **Loan officers** — need a risk score with a reason, not just a number.
- **Compliance/ops staff** — need fast, grounded answers to policy questions and clear explanations for flagged cases.

- **Risk/fraud teams —** need a system that catches fraud without drowning them in false positives.

Scope

In scope: credit risk scoring, transaction fraud detection, KYC document classification/verification, a RAG-based compliance assistant with tool-calling into the other three models, containerized deployment with monitoring. **Out of scope:** real payment processing, actual regulatory certification/compliance sign-off, live production customer data (this project uses public datasets only).

Success criteria

- Credit model: strong ROC-AUC on held-out data, with SHAP explanations for every prediction.
- Fraud model: meaningfully higher recall than the classical baseline at a controlled false-positive rate.
- Document module: correctly classifies document type and extracts key fields with high accuracy on test images.
- Assistant: answers grounded in retrieved documents (measured against a small hand-built eval set), correctly invokes the right tool when asked applicant- or transaction-specific questions.
- Platform: all four services running as separate containers behind one gateway, with at least one drift-monitoring dashboard live.

Assumptions & constraints

- Built entirely on public/synthetic datasets — no real customer data.
- Deployed on free-tier cloud infrastructure; designed to demonstrate production patterns at small scale, not handle production-scale traffic.
- Built solo, over roughly 4–5 months, in parallel with learning each underlying topic.